

25	In a velocity selection, positively charged particles $+q$ enter an electric field of strength E and a magnetic field of flux density B that are applied perpendicularly to each other in the same region. Particles which are travelling at speed v are undeflected by the cross-fields. Which of the following is necessarily true?
	A To select particles of charge $-q$ with speed v, there is a need to reverse the direction of either E or B.
	B To select charged particles with speed $2v$, with the magnitude of B unchanged, the magnitude of E needs to be doubled.
	C For particles of charge $-q$, there is a need to reverse the directions of both E and B in order to select the particles with speed v.
	D If the magnitude of charge doubles, the magnitudes of the electric field strength and magnetic flux density cannot remain as E and B to select the particles with speed v.

Answer: B

For the charged particles to be undeflected by the crossed-fields,

$$F_E = F_B$$

$$qE = Bqv$$

$$v = E/B$$

Option A – False. If the charge is of opposite polarity, with no change made at all to E and B , the direction of both forces reverse, equilibrium of forces to select those with speed v is still achieved.

Option B – True. Since $v = E/B$.

Option C – False. Because with directions of E and B maintained, particles of speed v can still be selected.

Option D – False. If q doubles, F_E and F_B increase by the same factor. q has no effect on the selected speed.